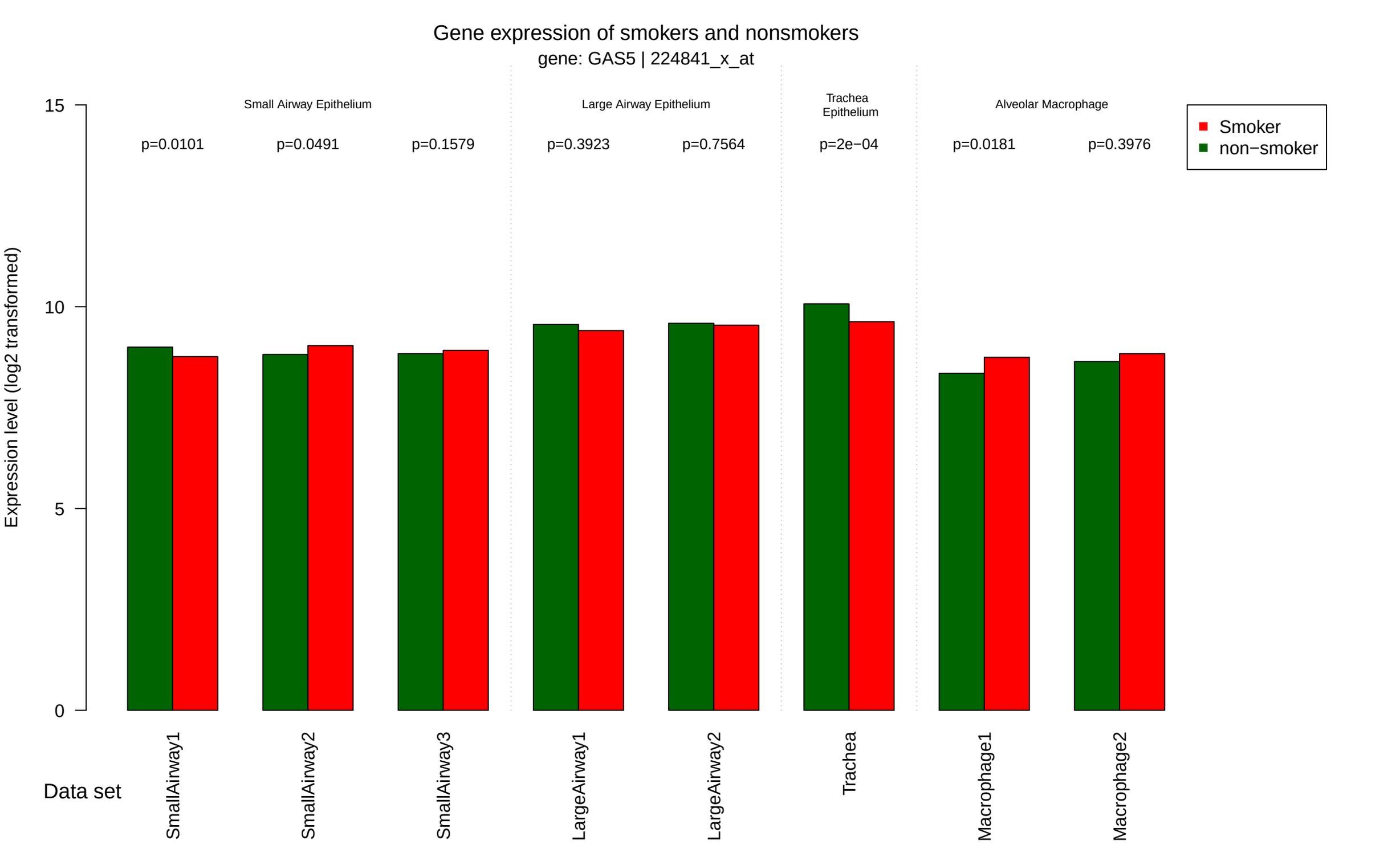
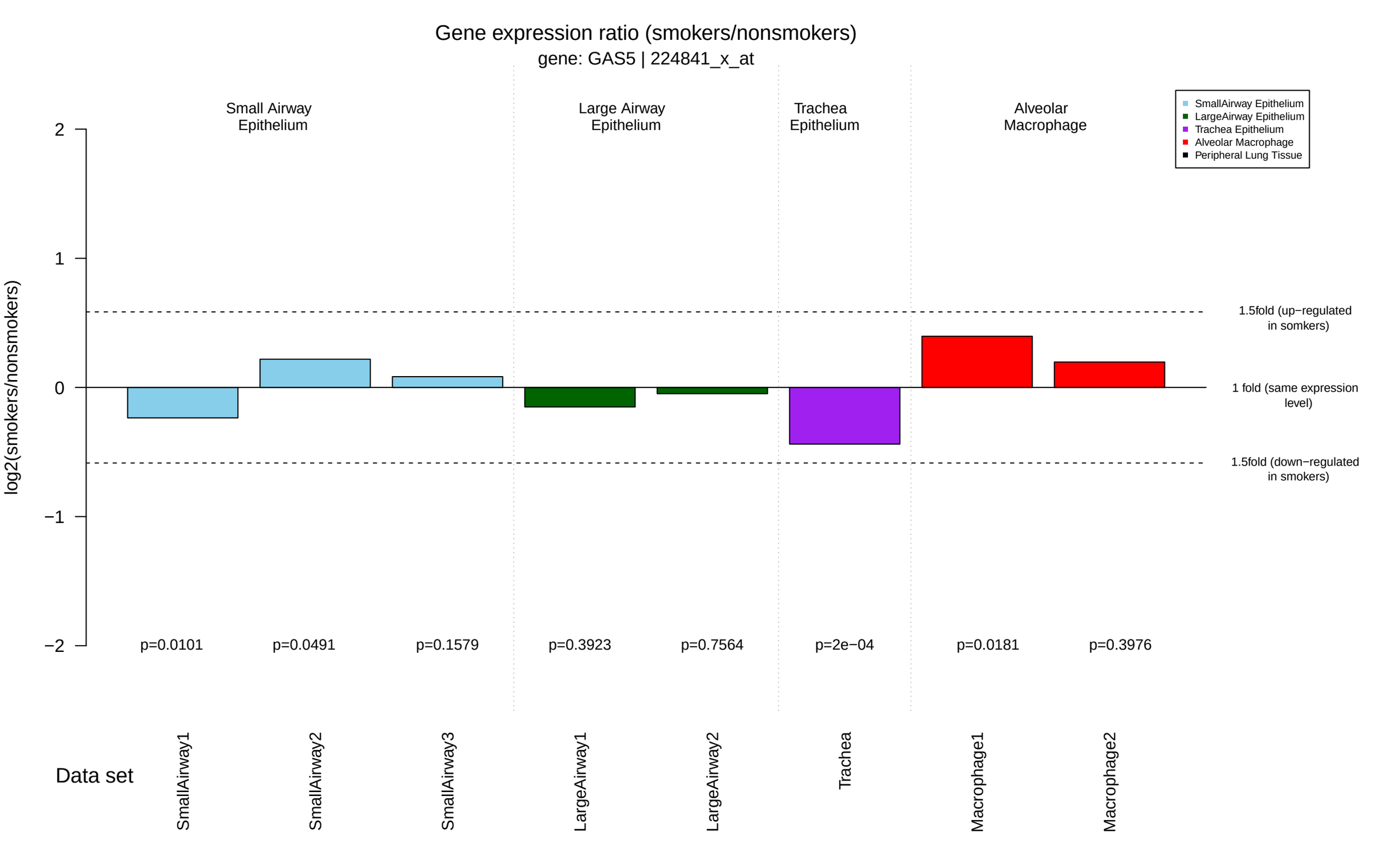
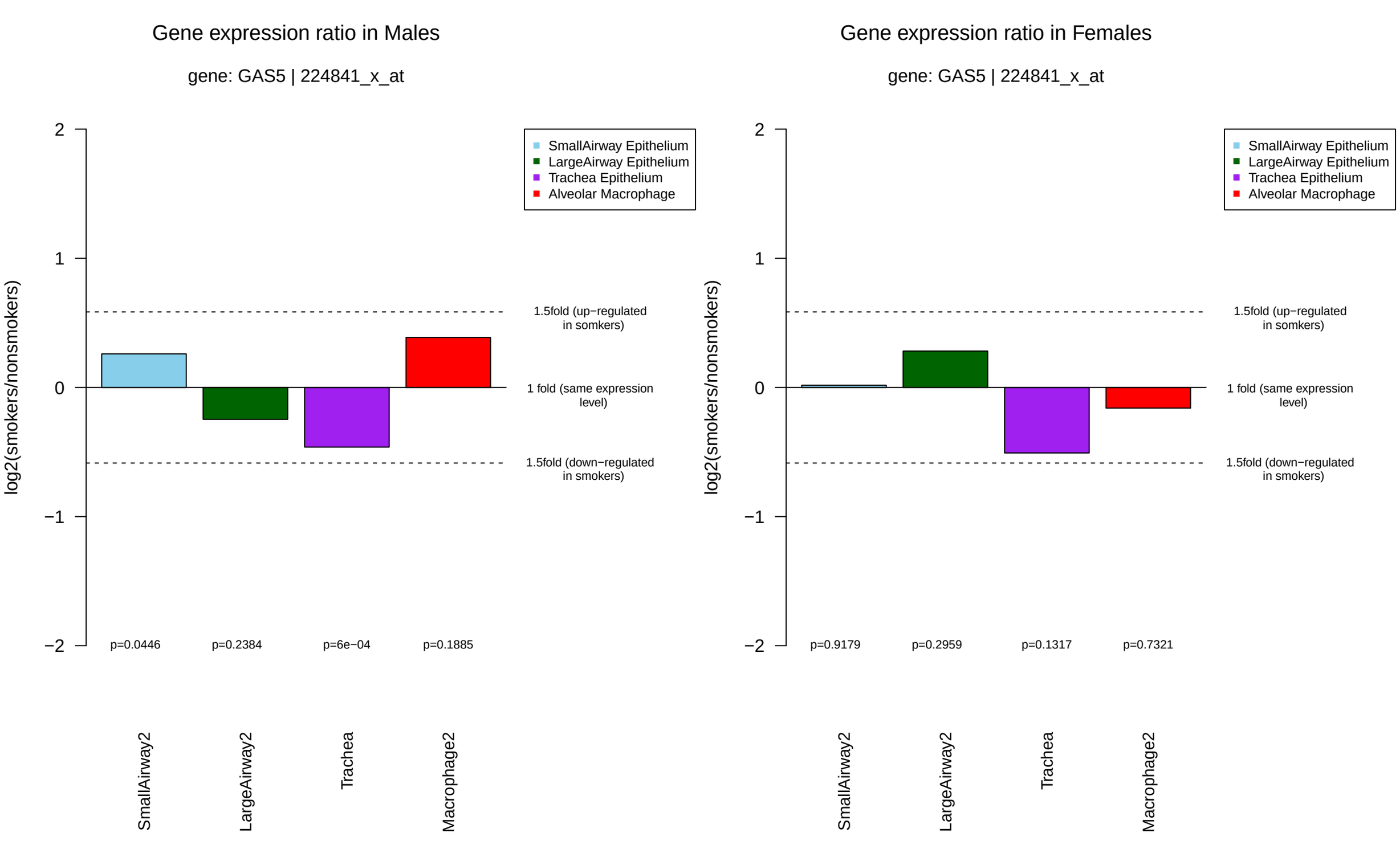
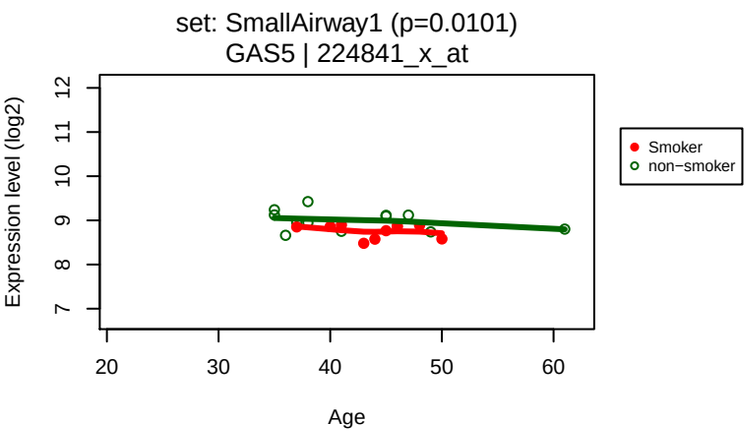
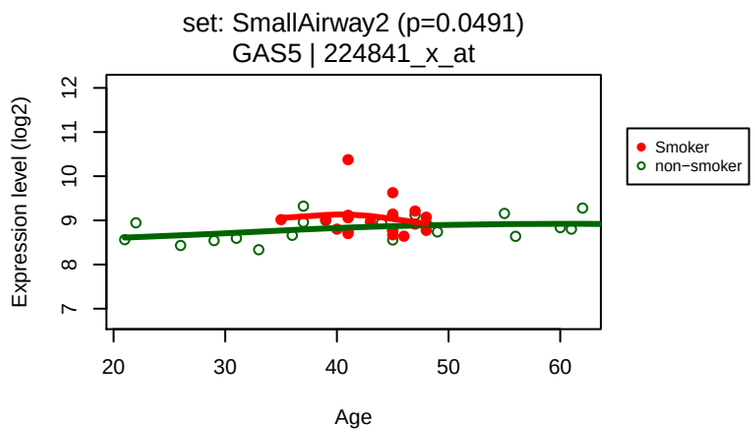
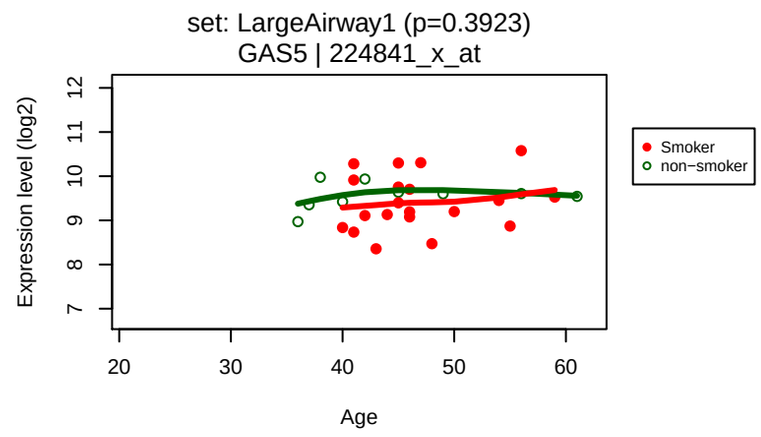

Gene expression of smokers and nonsmokers
gene: GAS5 | 224841_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0101
p=0.0491
p=0.1579
p=0.3923
p=0.7564
p=2e−04
p=0.0181
p=0.3976
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

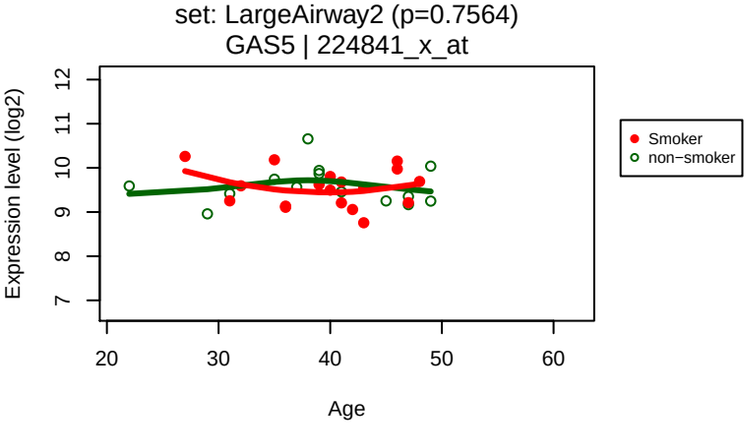

Gene expression ratio (smokers/nonsmokers)
gene: GAS5 | 224841_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0101
p=0.0491
p=0.1579
p=0.3923
p=0.7564
p=2e−04
p=0.0181
p=0.3976
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

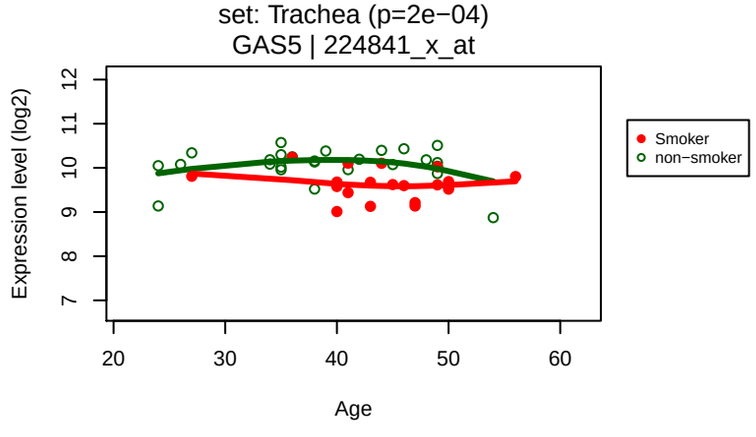

Gene expression ratio in Males
gene: GAS5 | 224841_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0446
p=0.2384
p=6e−04
p=0.1885
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GAS5 | 224841_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9179
p=0.2959
p=0.1317
p=0.7321
SmallAirway2
LargeAirway2
Trachea
Macrophage2

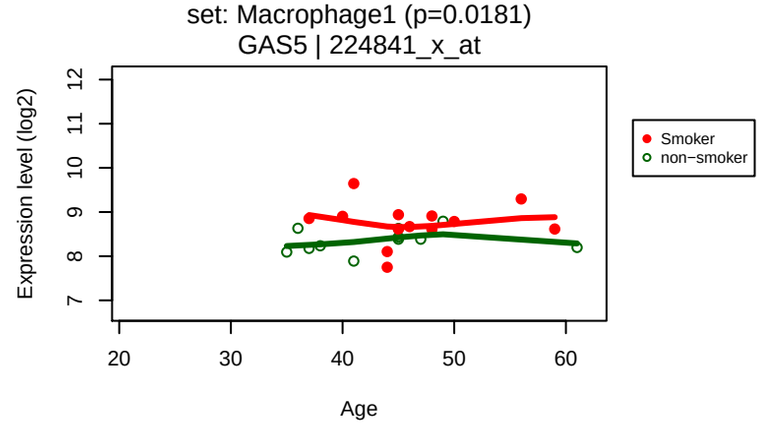

set: SmallAirway1 (p=0.0101)
GAS5 | 224841_x_at
Expression level (log2)
Age
Smoker
non−smoker

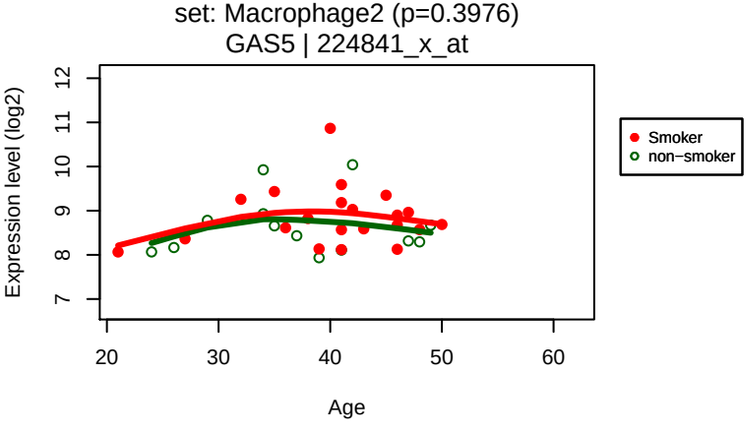

set: SmallAirway2 (p=0.0491)
GAS5 | 224841_x_at
Expression level (log2)
Age
Smoker
non−smoker

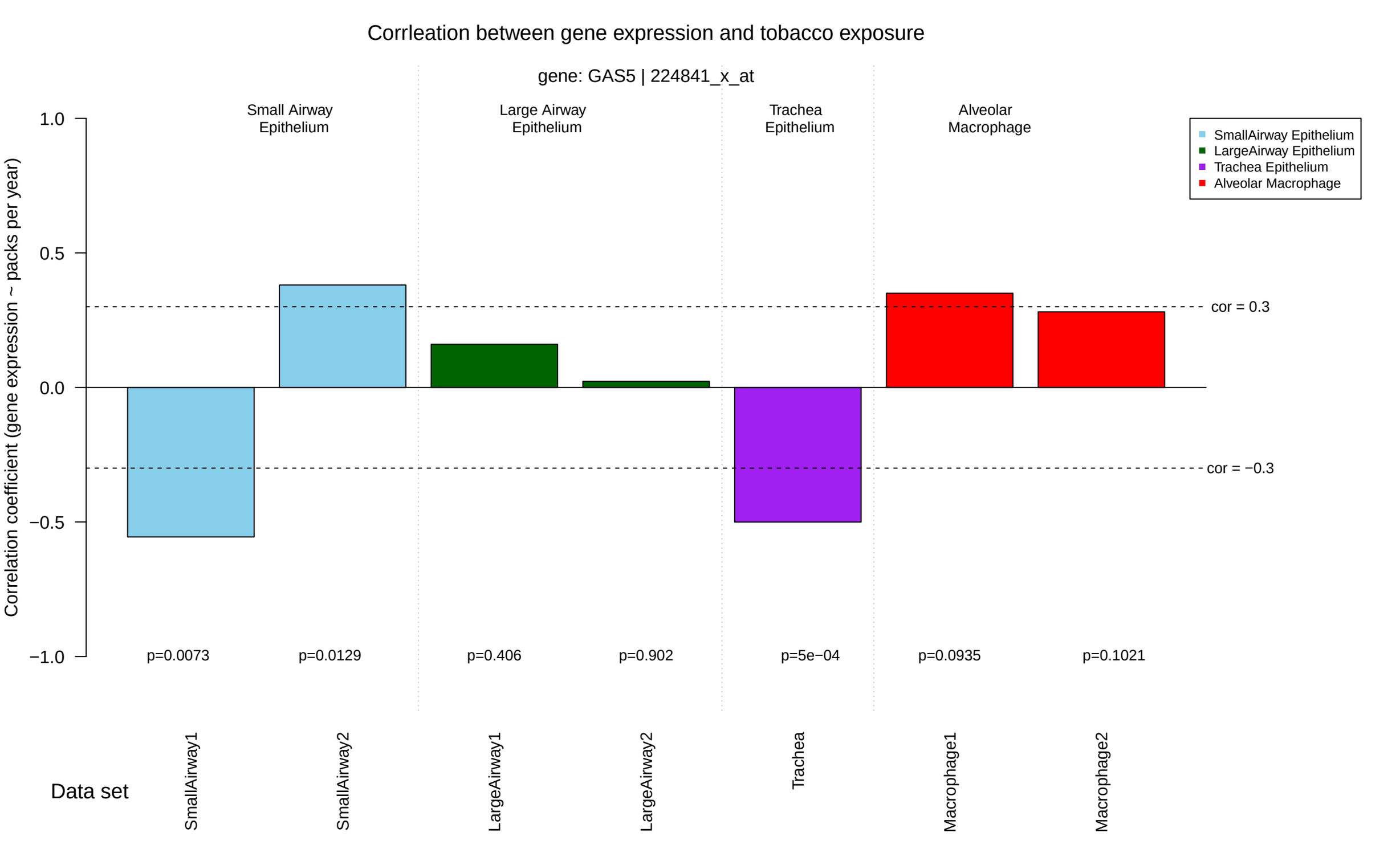

set: LargeAirway1 (p=0.3923)
GAS5 | 224841_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7564)
GAS5 | 224841_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=2e−04)
GAS5 | 224841_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0181)
GAS5 | 224841_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3976)
GAS5 | 224841_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GAS5 | 224841_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0073
p=0.0129
p=0.406
p=0.902
p=5e−04
p=0.0935
p=0.1021
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2